# LOT-PLATE-Builders-Manual

# LOT PLATE — Plan (HW-01)

CODENAME: LOT PLATE (P1) · OWNER: S-2 Vadik Marmeladov (Inventor, COSMO® CIA) · DATE: 2026-10-31 · REV A STATUS: PLAN. Nothing here is built or tested. All prices/dimensions marked ESTIMATE are unverified.

## 1. What it is

A 40 x 40 mm flat stainless-steel “plate” computer, ~5 mm target thickness, bound to one LOT account. It receives autonomous notifications from lot-systems.com (pager-style, e.g. “Coffee time!”), shows them on a small screen, reads the room (weather/air/motion/light), has a camera, and one button. The button is **COPY**: press once and a signal returns to the site’s **Log tab** (a device_copy log entry). It is the first physical precursor of COSMO® hardware (COSMO® target 2028–2029, see About.tsx “COSMO®”; requires Purple Benchmark Arbitrage™). It is NOT COSMO®.

## 2. Requirement trace (S-2 list → decision)

| # | Requirement | Decision | Doc |
|---|---|---|---|
| 1 | PCBWay | PCB + SMT assembly + CNC 316L body all quoted at PCBWay | 08 |
| 2 | Pager-like notification from AI site | Device polls/pushes /api/device/*; site AI (existing job engine) enqueues notifications | 07 |
| 3 | 2-part stainless body | Front bezel + back plate, 316L, 4 screws or laser-weld seam (DVT) | 04 |
| 4 | Flat silver square 4x4 cm x 5 mm | 40x40 mm, 5.0 mm target; EVT at ≤7.5 mm (see risk R1) | 04 |
| 5 | Camera | Thin DVP module on ESP32-S3; still capture + on-demand upload | 05 |
| 6 | LOT API connector | Device token + REST (/api/device/*), reference sim in tools/ | 07 |

| # | Requirement | Decision | Doc |
|---|---|---|---|
| 7 | PDF manuals | Generated from these .md → manuals/*.pdf | 09 |
| 8 | Compress info each session | STATE.md (≤1 page) rewritten each session; reports verbatim in docs/benchmark | README |
| 9 | Firmware documents | 06-FIRMWARE.md + firmware/ skeleton | 06 |
| 10 | Software to connect with firmware | tools/lot-device-sim.mjs + serial provisioning spec | 06/07 |
| 11 | Separate documents | One file per concern (this folder) | — |
| 12 | Charger | Qi dock (off-the-shelf EVT, custom puck PVT) | 05 |
| 13 | 100 units run | Cost + schedule model | 08 |
| 14 | Weather sensor | Bosch BME688 (T/RH/P/gas) | 05 |
| 15 | AI-grade off-the-shelf sensors | BME688 (BSEC AI gas classes), BMI270/BHI260 class IMU, VEML7700 light; ESP32-S3 vector instr. for on-device TinyML | 05 |
| 16 | Button “Copy” → Log tab | POST /api/device/event {type:"copy"} → Log event:"device_copy" | 07 |
| 17 | One side polished stainless | Back plate, mirror polish (Ra ≤0.05 µm target, ESTIMATE) | 04 |
| 18 | Other side camera+screen+button | Front bezel | 04 |
| 19 | Wireless charger | Qi RX (BQ51050B class) + glass window in back plate | 05 |

## 3. Three physics problems S-2 must decide on (the real plan risk)

These are why a sealed all-steel plate does not work as drawn. Detail in 04. 1. **RF**: a closed stainless shell is a Faraday cage; Wi-Fi/BLE die. Fix: non-metal aperture (ceramic/PEEK isolator along one seam edge + glass window). 2. **Qi**: steel blocks inductive charging (eddy currents). Fix: non-conductive window over the coil (⌀~26 mm glass/sapphire disc set flush in the polished back, mirror-black look). Fallback: 2 magnetic pogo pads (violates req 19 — S-2 call). 3. **Thickness**: 5.0 mm with camera + screen + battery + steel walls does not stack

with an ESP32 module. EVT builds thicker; 5.0 mm needs bare-die ESP32-S3 + ≤1.6 mm cell + ≤2.3 mm camera. Decision needed: is 5 mm a hard spec or DVT goal?

## 4. Architecture (one paragraph)

Site AI (existing QIE/jobs) → notification row in DB → GET /api/device/notifications (long-poll 30 s, later MQTT/SSE) → ESP32-S3 shows text, vibrates/beeps (optional haptic, EVT-2) → button “COPY” → POST /api/device/event → Log tab entry visible to the operator at lot-systems.com. Sensors upload a compressed reading every 5 min (POST /api/device/telemetry); camera uploads only on explicit request. Everything the device does is written to the Log (audit-by-default, matches LOT-NODE-0 principle “record every act”).

## 5. Non-goals (v1)

No cellular, no on-device LLM, no always-on video, no mic (privacy; add only with S-2 sign-off), no COSMO® behavioral-profile transfer (Soul Sync Protocol™ is a later gate).

## 6. Open questions for S-2

Q1 5 mm hard or goal? · Q2 glass back window acceptable (R2)? · Q3 screen: color IPS (baseline) vs e-paper (battery)? · Q4 haptic/buzzer wanted? · Q5 camera privacy shutter? · Q6 target market for 100 units (internal/beta/sale → certification scope)? · Q7 per-user or per-device account binding?

# LOT PLATE — Components Buying List (HW-02)

REV A · 2026-10-31 · **Sandbox had no access to vendor sites: no live prices, stock, or SKU pages were checked.** Links are manufacturer / distributor entry points. Prices are ESTIMATE (USD, qty-100 per unit). A human must verify stock + exact MPN on DigiKey / Mouser / LCSC before ordering. Mark each line VERIFIED in this file when done.

## A. EVT electronics (per unit)

| Ref | Part (MPN class) | Function | Est $ | Link |
|---|---|---|---|---|
| U1 | Espressif ESP32-S3-MINI-1-N8 (pre-certified module; DVT | MCU, Wi-Fi, BLE, DVP camera | 3.5 | https://www.espressif.com/en/products/modules |

| Ref | Part (MPN class) | Function | Est $ | Link |
|---|---|---|---|---|
| | may go bare ESP32-S3FN8) | | | |
| U2 | Bosch BME688 | Weather + AI gas (T/RH/P/VOC) | 8.0 | https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme688/ |
| U3 | Bosch BMI270 (or BHI260AP for on-chip AI) | IMU, tap/gesture, step | 3.0 | https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/ |
| U4 | Vishay VEML7700 | Ambient light (auto-brightness) | 1.5 | https://www.vishay.com/en/optical-sensors/ |
| U5 | TI BQ51050B (Qi RX + Li-ion charger, CSP) | Wireless charging | 3.0 | https://www.ti.com/product/BQ51050B |
| U6 | Analog Devices MAX17048 | Fuel gauge | 1.5 | https://www.analog.com/en/products/max17048.html |
| U7 | 3.3 V buck-boost/LDO (e.g. TI TPS62840 / TPS7A02) | Rail | 1.0 | https://www.ti.com/power-management/overview.html |
| L1 | Qi RX coil, thin ≤0.6 mm, ø~25 mm, with ferrite (Würth Elektronik WE-WPCC thin series class) | Wireless power | 2.5 | https://www.we-online.com/en/components/products/WE-WPCC |
| BT1 | Thin LiPo, target ≤1.6 mm, 50–100 mAh, with protection IC (custom-spec from cell house; quote at PCBWay-adjacent vendors / Grepow / Great Power) | Power | 4.0 | https://www.digikey.com/en/products/filter/batteries-rechargeable-secondary |

| Ref | Part (MPN class) | Function | Est $ | Link |
|---|---|---|---|---|
| DS1 | 1.3″ IPS 240x240 ST7789 FPC panel (baseline) — ALT: 1.54″ e-paper GDEY0154D67 | Screen | 4.0 | https://www.buydisplay.com (IPS) · https://www.good-display.com (e-paper) |
| CAM1 | OmniVision OV2640-class 2 MP FPC module, thinnest available (need ≤2.5 mm; DVT: OV2640 CSP bare) | Camera | 4.5 | https://www.ovt.com/products/ |
| SW1 | Low-profile tactile / metal dome ≤0.6 mm travel (e.g. Alps SKRP / C&K PTS) | COPY button | 0.5 | https://tech.alpsalpine.com/e/products/category/tactile-switches/ |
| LS1 | (optional) 0603 piezo / LRA coin ≤2.0 mm | Pager alert | 1.5 | https://www.digikey.com/en/products/filter/haptic-actuators |
| PCB | 4-layer, 0.8 mm, ENIG, 38x38 mm, antenna keep-out | Board | see 08 | https://www.pcbway.com/ |
| misc | Passives, ESD (USBLC6), pogo test pads, FPC conn., flex ant. | — | 4.0 | https://www.lcsc.com/ |
| | **Electronics subtotal (excl. PCB/SMT)** | | **≈ 42.5** | |

# B. Mechanical

| Item | Spec | Est $ /unit @100 | Link |
|---|---|---|---|
| Front bezel | 316L, CNC, camera + screen + button apertures | 20–35 | https://www.pcbway.com/cnc-machining-quote.html |
| Back plate | 316L, CNC, mirror polish, glass-window pocket | 20–35 | same |
| Back window | Glass/sapphire disc ⌀26 x 0.5, AR-free, black-backed | 3–8 | https://www.pcbway.com/cnc-machining-quote.html (or specialty glass cutter) |
| Seam isolator | PEEK / zirconia ring-segment (RF aperture) | 2–4 | PCBWay CNC / 3D |
| Screws | M1.0 x 2.5 torx, 316 | 0.4 | https://www.mcmaster.com |
| Gasket / adhesive | thin PSA, IPX1 target | 0.5 | — |
| Cover glass | 0.5 mm chem-strengthened, front | 2–4 | — |

## C. Charger / tools

| Item | Spec | Est $ | Link |
|---|---|---|---|
| EVT dock | Any 5 W Qi pad, thin (test only) | 10 | https://www.digikey.com/en/products/filter/wireless-charging-transmitters |
| PVT dock | Custom puck: Qi TX IC (e.g. TI BQ500212A class) + magnet ring, USB-C | 12–18 | https://www.ti.com/product/BQ500212A |
| Programmer | ESP-Prog or USB-C native (S3 has USB) | 0 (pads) | — |
| Test | Pogo-pin jig (PCBWay) | 150 once | https://www.pcbway.com/ |
| Dev kit (P1) | ESP32-S3-EYE or XIAO ESP32S3 Sense (cam) + BME688 breakout + e-paper breakout | ~90 once | https://www.seeedstudio.com/XIAO-ESP32S3-Sense-p-5639.html |

## D. Roll-up per unit @100 (ESTIMATE)

Electronics 42.5 + PCB/SMT 14–20 + steel/glass/isolator 46–86 + battery incl. above + dock 12–18 + packaging/QA 8 ≈ **$125–175 landed, before certification/NRE**. See 08.

# LOT PLATE — Roadmap & Analysis (HW-03)

REV A · 2026-10-31 · Dates are ESTIMATE from today; one-person-plus-AI cadence assumed. Gate = go/no-go for S-2.

| Phase | Window | Output | Gate (exit criteria) |
|---|---|---|---|
| P0 Plan | Oct–Nov 2026 | These docs, S-2 answers to Q1–Q7 | R1–R3 decisions made |
| P1 Breadboard | Nov–Dec 2026 | XIAO ESP32S3 Sense + BME688 + screen; firmware skeleton talks to server; **server /api/device/* shipped** (Ship Mode) | Notification appears on screen ≤30 s; COPY writes a Log entry; 24 h soak |
| P2 EVT-1 PCB | Jan–Feb 2027 | 10 boards PCBWay (assembled), 3D-printed/Al body | Boots, Wi-Fi, all sensors read, camera frame, Qi charges battery |
| P3 EVT-2 steel | Mar 2027 | 10 x 316L two-part bodies + glass window + isolator | **RF + Qi measured through real steel** (go/no-go for design) |
| P4 DVT | Apr–Jun 2027 | 30 units, 5.0 mm attempt (bare S3, thin cell), jig, drop/thermal | ≥90 % yield; battery ≥5 days @ 10 notif/day; pre-compliance scan |
| P5 PVT | Jul 2027 | 100-unit pilot build drawing from frozen BOM | Cpk on polish/fit; UN38.3 + FCC/CE plan done |
| P6 Run | Aug–Sep 2027 | 100 units shipped | RMA < 5 % in 30 days |

## Critical path

Steel-window RF/Qi result (P3) → everything. Pull a cheap proof forward: in P1 buy a 40x40 mm 316L blank with a pocket (PCBWay CNC, ~$60 x 3) and test Wi-Fi RSSI + Qi with a Qi RX dev board **before** committing PCB layout.

## Risk register

| ID | Risk | L | I | Mitigation |
|---|---|---|---|---|
| R1 | 5 mm stack infeasible | H | H | Stack budget in 04; decide EVT 7.5 / DVT 6 / goal 5 |
| R2 | Steel kills RF | H | H | Isolator seam + glass window; external flex antenna on seam edge; test P1 |

| ID | Risk | L | I | Mitigation |
|---|---|---|---|---|
| R3 | Steel blocks Qi | H | H | Glass window over coil; fallback pogo |
| R4 | Thin LiPo supply / safety (UN38.3, swelling) | M | H | Use protected cell from certified vendor; no custom cell for 100 units if MOQ high |
| R5 | Camera ≤2.5 mm sourcing | M | M | OV2640 CSP flex module; or drop camera on rev B |
| R6 | Polish scratching in handling | M | L | Film + QA gloves; PVD optional |
| R7 | Certification (FCC/CE) cost if sold | M | M | Pre-certified module; modular approval needs intact antenna, may need re-test in metal (≈$8–15k ESTIMATE) |
| R8 | Privacy (camera) | M | H | Shutter or LED + log every capture to Log tab; no auto-upload |
| R9 | Server abuse via device tokens | M | M | Scoped tokens, rate limit, revoke in Settings |
| R10 | Scope: COSMO® confusion | L | M | Brand line in About: PLATE ≠ COSMO® |

## Analysis (honest)

- Software side is low risk and fast (P1 in ~4 weeks). Hardware side is dominated by the steel enclosure, not the electronics.
- Cost drivers at 100 units: CNC steel (≈40 %) and polish. Doubling qty to 200 cuts per-unit steel ≈15 % (ESTIMATE).
- Cheapest way to cut risk: build P1 first, then a “steel test coupon” — both < $300.
- Schedule slack: none for certification if units are sold; internal/beta units avoid it.

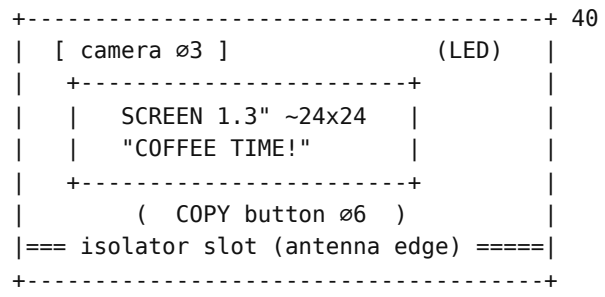
# LOT PLATE — Mechanical & RF (HW-04)

REV A · 2026-10-31 · ESTIMATE throughout; needs CAD + bench validation (P1/P3).

## Body

- 2 parts, 316L stainless: **FRONT** (bezel: screen window, camera hole ø3, button) and **BACK** (mirror-polished plate).
- 40.0 x 40.0 mm, corner R3. Seam: 4 x M1.0 torx (EVT/DVT); laser-weld not used (blocks service + would defeat isolator).
- One edge carries a 2 x 12 mm **PEEK/zirconia isolator** — the RF slot aperture for the antenna.
- Back: ø26 mm pocket 0.5 mm deep, glass disc bonded flush = Qi window. Polished steel around it.

## Front layout (top view, mm)




Camera, screen and button are side-by-side, not stacked, so only the tallest single item sets thickness.

## Stack budget (Z, mm) — ESTIMATE

| Layer | EVT (module) | DVT (bare die) |
|---|---|---|
| Back plate steel | 0.8 | 0.5 |
| Glass window region (same Z) | — | — |
| Battery | 3.0 | 1.6 |
| PCB | 1.0 | 0.8 |
| Tallest component (module 2.4 / QFN 0.9) | 2.4 | 0.9 |
| Display + FPC | 1.8 | 1.05 (e-paper) |
| Front cover glass | 0.5 | 0.4 |
| Gaps/adhesive | 0.3 | 0.2 |
| **Stack if all stacked** | **~9.8** | **~5.5** |

Battery under PCB, display over PCB; camera/button in the side strip (not additive). Realistic EVT ≈ 7.5–8 mm, DVT ≈ 5.5–6 mm. **5.0 mm is a stretch goal, not a promise.**

## RF

- Stainless 316L is conductive; sealed → near-zero Wi-Fi. Antenna must radiate through the isolator slot / glass.
- Use module with PCB antenna at the isolator edge, ≥6 mm keep-out of copper/steel inside; or u.FL → flex antenna on the slot (preferred for

steel).
- Test (P1 coupon, P3 real): RSSI @ 1 m vs open-air; accept ≥ −10 dB loss; BLE pairing range ≥3 m.

## Qi

- Coil ⌀~25 under the glass window; ferrite sheet behind coil (between coil and PCB). Coil-to-steel gap ≥1 mm, otherwise efficiency collapses.
- Test: 5 W pad, charge current ≥100 mA at 1 mm air gap; thermal <40 °C.
- Dock magnet ring aligns plate; magnet ≠ near compass/IMU (calibrate offset).

## Finish / ingress

Back: mirror polish, optional clear PVD. Front: bead-blast bezel. IPX1 target (drip), not waterproof.

# LOT PLATE — Electronics (HW-05)

REV A · 2026-10-31 · schematic not drawn yet; this is the block + net plan for the PCBWay design.

## Block diagram

```
Qi coil -> BQ51050B -> LiPo -> MAX17048 (I2C)
                        |-> 3V3 reg -> ESP32-S3 -+- I2C0: BME688, BMI270, VEML7700, MAX17048
                                                +- SPI : display (ST7789)
                                                +- DVP : camera (8-bit, XCLK 20 MHz, SCCB on I2C1)
                                                +- GPIO: COPY button (wake, ext0), haptic, status
LED
                                                +- USB : native USB-C pads (flash/log) / pogo jig
```

## Pin plan (ESP32-S3-MINI-1, EVT; confirm vs datasheet, avoid strap pins 0/3/45/46, PSRAM-less N8)

| Signal | GPIO | Note |
|---|---|---|
| I2C0 SDA/SCL | 8 / 9 | sensors + gauge, 400 kHz |
| SPI SCK/MOSI/CS/DC/RST | 12/11/10/13/14 | display |
| BTN_COPY | 4 | RTC-capable wake |
| HAPTIC | 5 | PWM |
| CAM D0–D7, PCLK, VSYNC, HREF, XCLK | per Espressif ESP32-S3-EYE ref | use their reference table |
| CHG_INT / GAUGE_ALRT | 6 / 7 | |
| LED | 15 | |

Camera DVP eats ~12 GPIOs; with N8 (no PSRAM) a JPEG frame at QVGA/VGA fits; use N8R2 if SVGA wanted.

## Power

Cell 3.0–4.2 V. Sleep: ESP32 deep sleep ~10 µA + sensors ~10 µA. Wake every 30 s for poll via light sleep, or use MQTT keepalive (EVT-2). Budget ESTIMATE: avg ≈ 1.2 mA → 80 mAh ≈ 2.7 days; target 5 days needs duty-cycle (poll 60 s) + display only on event. Screen stays off by default (pager behavior), wakes on notification/button.

## AI-grade sensors (what “AI” means here, honestly)

- BME688: Bosch BSEC library classifies gas patterns (needs BME AI-Studio training; closed-source lib, licence terms to check).
- BMI270/BHI260: on-chip gesture/tap; BHI260AP runs Bosch AI-ready firmware.
- ESP32-S3: TinyML (ESP-NN, TFLite-Micro) for tap/room-state; no cloud LLM on device.
- Camera: capture only; vision AI runs server-side via existing AI engine on explicit request.

## PCBWay design rules (ESTIMATE spec for quote)

4 layers, 0.8 mm, 38x38 mm, ENIG, 0.1/0.1 mm trace/space, 0.2 mm via, impedance on RF trace (50 Ω), antenna keep-out zone, castellated test pads, fiducials, panelized 2x4 for SMT. Files: Gerber RS-274X + BOM + CPL + assembly drawing.

## Charger

Qi RX BQ51050B → charge 100 mA CC/CV to 4.2 V. Dock: 5 W Qi TX, magnet ring. USB-C on dock only; plate has none (pads under glass for service).

# LOT PLATE — 100-Unit Run: PCBWay & Cost (HW-08)

REV A · 2026-10-31 · All figures ESTIMATE; no live quotes obtained (sandbox offline to vendors). Get real quotes in P2.

## PCBWay scope

| Service | Use | Link |
|---|---|---|
| PCB fab | 4L 0.8 mm ENIG | https://www.pcbway.com/ |
| PCB assembly (turnkey/consigned) | SMT both sides, X-ray BGA/QFN check | https://www.pcbway.com/pcb-assembly.html |

| Service | Use | Link |
|---|---|---|
| CNC machining | 316L front/back, polish | https://www.pcbway.com/cnc-machining-quote.html |
| 3D printing (EVT-1 body) | SLA/MJF | https://www.pcbway.com/rapid-prototyping/ |

Upload: Gerbers, BOM (MPN + LCSC/DigiKey #), CPL, drawings (STEP + PDF with tolerances ±0.05, Ra callout). Ask for DFM review before P2.

## Cost model (qty 100, USD)

| Line | Per unit | x100 |
|---|---|---|
| Electronics BOM (02-A) | 42.5 | 4,250 |
| PCB + SMT + test (ESTIMATE 14–20) | 17 | 1,700 |
| Steel x2 + polish + glass + isolator (46–86) | 66 | 6,600 |
| Dock (custom puck) | 15 | 1,500 |
| Packaging + QA + shipping in | 8 | 800 |
| **Unit subtotal** | **148.5** | **14,850** |
| NRE: stencil/jig/fixtures/CAD review | | ~1,500 |
| Pre-compliance + UN38.3 cell docs | | ~3,000 |
| FCC/CE formal (only if sold; ESTIMATE 8–15k) | | 0–15,000 |
| Prototype iterations P1–P4 (10+10+30 units) | | ~6,000 |
| **Run total** | | **≈ $25k internal / ≈ $40k sellable** |

Spare 10 % yield buffer (110 starts) adds ≈ $1.5k.

## Quality & shipping

Incoming: sample AQL on steel (flatness, Ra, aperture position). EOL jig: flash, sensors, Wi-Fi ping to test server, button, Qi, camera. Serial + QR (hwId). Batteries: UN38.3 transport, ship by ground/sea or certified air with <100 Wh, packed per PI 967/966.

## Schedule (see 03): order P6 materials Jul 2027, ship Sep 2027 (lead time CNC 2–3 wk, PCB 2 wk, assembly 2 wk, ESTIMATE).